

Interview Questions — Python - DB and Python Framework

Total Questions: 63

1. What is Django and why is it popular for web development?

Hint: Think about rapid development and built-in features.

2. Explain the main differences between Django and Flask.

Hint: Focus on project structure and how much is provided out-of-the-box.

3. What is the MVT architecture in Django?

Hint: Describe the roles of Model, View, and Template.

4. How do you install Django and start a new project?

Hint: Mention the command-line tools involved.

5. After creating a new Django project, what are the main files and folders you see in the project structure?
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6. What is the difference between a Django project and a Django app?

Hint: One is the whole site, the other is a reusable component.

7. How do you map a URL to a view in Django?

Hint: Think about the `urls.py` file.

8. What is the purpose of the `settings.py` file in a Django project?
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9. Write a simple Django view that returns 'Hello, World!' as an `HttpResponse`.
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10. How would you create a new Django app called 'accounts' and register it with your project?
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11. What is the purpose of the `templates` folder in a Django app?
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12. Explain template inheritance in Django and its benefits.

Hint: Think about avoiding repetition across pages.

13. How do you load static files like CSS or JavaScript in a Django template?

14. How would you add Bootstrap to your Django project using a CDN?

15. What is a Django model and why do we use it?

Hint: Consider how Django interacts with the database.

16. How do you create a model for a 'Doctor' with fields for name and specialization?

17. What is the process to apply changes to your Django models in the database?

Hint: Mention the commands used to create and apply migrations.

18. How would you connect a Django project to a MySQL database instead of the default SQLite?

19. What steps are needed to make a new model appear in the Django admin interface?

20. How can you customize the admin list display for a model in Django?

21. How would you add a filter for specialization in the Django admin for the Doctor model?

22. What is the difference between a Django Form and a ModelForm?

Hint: Think about how each form is created and what it is used for.

23. Explain the difference between GET and POST requests in the context of Django forms.

24. Why is the CSRF token important when handling form submissions in Django?

Hint: Consider security for web forms.

25. Write a Django Form class to add a new Doctor with fields for name and phone number.

26. How would you display validation errors in a Django form template?

27. How does a ModelForm help in saving data to the database?

Hint: Think about automatic field mapping.

28. Write a view function that uses a ModelForm to save a new Doctor to the database.

29. How does Django handle user authentication (signup, login, logout) by default?

30. What is password hashing and why is it important in user authentication?

31. How would you display the logged-in user's username in the navigation bar of your Django site?

32. What steps are required to send an email in Django upon user signup?

33. How can you use HTML email templates when sending emails in Django?

34. Describe how Django sessions work and what they are used for.

Hint: Think about storing user data between requests.

35. How would you implement a forgot password feature using OTP and Django sessions?

36. What are MEDIA_ROOT and MEDIA_URL settings used for in Django?

37. How would you allow users to upload and update their profile pictures in a Django app?

38. Explain the steps to create basic CRUD operations (Add, View, Edit, Delete) for the Doctor model.

39. How would you implement a delete confirmation page before removing a Doctor record?

40. What is AJAX and how can it be used to delete a Doctor record in Django without refreshing the page?

41. How would you send a JSON response from a Django view?

42. Explain the use of filter(), exclude(), and annotate() methods in Django ORM.

43. What is the difference between select_related() and prefetch_related() in Django ORM?

Hint: Both are for optimizing database queries with related objects.

44. How would you paginate a list of doctors in Django?

45. How does the login_required decorator work in Django?

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46. Describe how you would implement role-based access control, such as separate dashboards for doctors and patients.
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47. How can you restrict access to certain admin pages based on user permissions or groups?
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48. How would you integrate Google Login (social login) into a Django project?
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49. Describe the steps to send an OTP via SMS using an API like Twilio in Django.
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50. What are the main steps to integrate a payment gateway (such as Stripe or Paytm) in a Django application?
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51. How would you handle the payment callback and update the appointment status after a successful transaction?
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52. Explain how to embed a Google Map showing a doctor's clinic location in a Django template.
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53. How would you search for doctors by distance using geolocation APIs in Django?
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54. What steps are required to deploy a Django project to PythonAnywhere and make it live?
- Hint:** Think about code upload, static/media files, and database migration.
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55. How do you configure static and media files for production deployment in Django?
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56. Describe the process of pushing your Django project to GitHub and why version control is important.
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57. How would you ensure your deployed Django site is working correctly after deployment?
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58. In your Doctor Finder project, how did you implement appointment booking and payment integration?
- Hint:** Describe the workflow from user action to payment confirmation.
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59. How did you use filters such as specialization and location in your doctor search feature?
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60. Describe how you handled user registration, OTP verification, and password reset in your capstone project.
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61. If you were to improve your Doctor Finder project, what features or optimizations would you add?
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62. How could you use AI tools like ChatGPT or GitHub Copilot to speed up Django development, and what would you check before using their code?

Hint: Think about code suggestions and the importance of reviewing for security or correctness.

63. How might you use AI to help with tasks like writing email templates or generating test data in a Django project?